B) optimize $Z = 6x_1^2 + 5x_2^2$
Subject to $x_1 + 5x_2 = 7$
$x_1, x_2 \geq 0$

$\Rightarrow$ we have lagrange's function
$L(x_1, x_2, \lambda) = 6x_1^2 + 5x_2^2 - \lambda(x_1 + 5x_2 - 7)$
obtain the partial derivative.

$$\frac{\partial L}{\partial x_1} = 0 \quad , \quad \frac{\partial L}{\partial x_2} = 0 \quad , \quad \frac{\partial L}{\partial \lambda} = 0$$

$$\frac{\partial L}{\partial x_1} = 12x_1 - \lambda = 0 \quad , \quad \frac{\partial L}{\partial x_2} = 10x_2 - 5\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x_1 + 5x_2 - 7 = 0$$

$$\lambda = 12x_1 \quad , \quad 10x_2 = 5\lambda \quad , \quad x_1 + 5x_2 - 7 = 0$$

$\cancel{\lambda = 2x_2}$ $\quad \therefore \lambda = 2x_2$

$\because \lambda = 2x_2$

$\therefore 2x_2 = 12x_1$

$x_2 = 6x_1$

put in $x_1 + 5x_2 - 7 = 0$

$x_1 + 5(6x_1) - 7 = 0$

$x_1 + 30x_1 - 7 = 0$

$31x_1 = 7$

$$\boxed{x_1 = \frac{7}{31}}$$

put $x_1 = \frac{7}{31}$ in $x_2 = 6x_1$

$x_2 = 6\left(\frac{7}{31}\right)$

$$\boxed{x_2 = \frac{42}{31}}$$

$\lambda = 2x_2$

$\lambda = 2\left(\frac{42}{31}\right)$

$$\boxed{\lambda = \frac{84}{31}}$$

$f(x_1, x_2) = 6x_1^2 + 5x_2^2$

$h(x_1, x_2) = x_1 + 5x_2 - 7$

$$\Delta = \begin{vmatrix} 0 & \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_2 \partial x_1} \\ \frac{\partial h}{\partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_2^2} \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 0 & 1 & 5 \\ 1 & 12 & 0 \\ 5 & 0 & 10 \end{vmatrix}$$

$= -310$

$\therefore$ Δ is negative, X_0 is minima.

hence, $x_1 = \frac{7}{31}$, $x_2 = \frac{42}{31}$

$Z_{min} = 6\left(\frac{7}{31}\right)^2 + 5\left(\frac{42}{31}\right)^2$

$Z_{min} = \frac{294}{31}$

(c) Solve the following NLPP by Kuhn Tucker condition
Maximize $z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$
Subject to $2x_1 + x_2 \le 5$
$x_1, x_2 \geq 0$

$\Rightarrow$ $f(x_1, x_2) = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$
$h(x_1, x_2) = 2x_1 + x_2 - 5$

condition for Kuhn Tucker

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 10 - 4x_1 - \lambda(2)$$
$$= 10 - 4x_1 - 2\lambda = 0 \quad \text{—} (1)$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 4 - 2x_2 - \lambda(1)$$
$$= 4 - 2x_2 - \lambda = 0 \quad \text{—} (2)$$

$$\lambda(2x_1 + x_2 - 5) = 0 \quad \text{—} (3)$$
$$2x_1 + x_2 - 5 = 0 \quad \text{—} (4)$$
$$\lambda, x_1, x_2 \geq 0$$

case I : $\lambda = 0$

$10 - 4x_1 = 0$
$4x_1 = 10$
$x_1 = \frac{10}{4}$
$\boxed{x_1 = \frac{5}{2}}$

$4 - 2x_2 = 0$
$2x_2 = 4$
$x_2 = \frac{4}{2}$
$\boxed{x_2 = 2}$

put x_1, x_2 in equ^n (4)
$2(x_1) + x_2 - 5$
$= 2\left(\frac{5}{2}\right) + 2 - 5$
$= 2 \neq 0$

This value doesnt satisfy equ^n (4), hence $\lambda = 0$ doesn't yeild a feasible sol^n so we reject this value case □ : $2x_1 + x_2 - 5 = 0$

$$2x_1 + x_2 = 5 \quad - (5)$$

$$10 - 4x_1 - 2\lambda = 0$$

$$4 - 2x_2 - \lambda = 0$$

$$10 - 4x_1 - 2\lambda + 4 - 2x_2 - \lambda = 0$$

$$10 - 2(2x_1 + x_2) - 3\lambda + 4 = 0$$

$$10 - 2(5) - 3\lambda + 4 = 0$$

$$-3\lambda + 4 = 0$$

$$\boxed{\lambda = \frac{4}{3}}$$

put in eq^n (1)

$$10 - 4x_1 - 2\lambda = 0$$

$$10 - 4x_1 - 2\left(\frac{4}{3}\right) = 0$$

$$4x_1 = 10 - \frac{8}{3}$$

$$4x_1 = \frac{22}{3}$$

$$\boxed{x_1 = \frac{11}{6}}$$

put $\lambda = \frac{4}{3}$ in eq^n (2)

$$4 - 2x_2 - \frac{4}{3} = 0$$

$$2x_2 = 4 - \frac{4}{3}$$

$$x_2 = \frac{4}{3}$$

this value satisfy the necessary condition

∴ the optimal sol^n is $x_1 = \frac{11}{6}$, $x_2 = \frac{4}{3}$

$$z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$$

$$= 10\left(\frac{11}{6}\right) + 4\left(\frac{4}{3}\right) - 2\left(\frac{11}{6}\right)^2 - \left(\frac{4}{3}\right)^2$$

$$\boxed{z = \frac{91}{6}}$$

(B) using the method of lagrange's solve the following NLPP.

optimize $z = 4x_1 + 8x_2 - x_1^2 - x_2^2$

subject to $x_1 + x_2 = 4$

$x_1, x_2 \geq 0$

$\Rightarrow$ $L(x_1, x_2, \lambda) = (4x_1 + 8x_2 - x_1^2 - x_2^2) - \lambda(x_1 + x_2 - 4)$

$\frac{\partial L}{\partial x_1} = 4 - 2x_1 - \lambda$ $\qquad$ $\frac{\partial L}{\partial \lambda} = x_1 + x_2 - 4.$

$\frac{\partial L}{\partial x_2} = 8 - 2x_2 - \lambda$

$4 - 2x_1 - \lambda = 0$ — (1), $8 - 2x_2 - \lambda = 0$ — (2) $x_1 + x_2 - 4 = 0$ — (3)

$\lambda = -2x_1 + 4$, $\lambda = 8 - 2x_2$

$\lambda = 4 - 2x_1 = 8 - 2x_2$

$\therefore 4 - 2x_1 = 8 - 2x_2$

$-2x_1 + 2x_2 = 8 - 4$

$2(-x_1 + x_2) = 4$

$-x_1 + x_2 = 2$

$x_2 = x_1 + 2$

$x_1 + x_2 = 4$

$x_1 + (x_1 + 2) = 4$

$2x_1 = 2$

$\boxed{x_1 = 1}$

$1 + x_2 - 4 = 0$

$\boxed{x_2 = 3}$

$\lambda = 4 - 2(1)$

$\boxed{\lambda = 2}$

$f(x_1, x_2) = 4x_1 + 8x_2 - x_1^2 - x_2^2$

$h(x_1, x_2) = x_1 + x_2 = 4$

$$\Delta = \begin{vmatrix} 0 & \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} \\ \frac{\partial h}{\partial 2} & \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 1 & 1 \\ 1 & -2 & 0 \\ 1 & 0 & -2 \end{vmatrix}$$

$$= 4$$

$\therefore$ Δ is positive, X_0 is maxima

hence $x_1 = 1$, $x_2 = 3$

$$z = 4(x_1) + 8x_2 - x_1^2 - x_2^2$$

$$= 4(1) + 8(3) - 1^2 - 3^2$$

$$\boxed{z = 18}$$

1(c) Solve the following NLPP by Kuhn Tucker

Maximize $Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$

subject to $2x_1 + 5x_2 \le 98$

$x_1, x_2 \ge 0$

$f(x_1, x_2) = 2x_1^2 - 7x_2^2 + 12x_1x_2$

$h(x_1, x_2) = 2x_1 + 5x_2 - 98$

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$4x_1 + 12x_2 - 2\lambda = 0$ — (1)

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$-14x_2 + 12x_1 - 5\lambda = 0$ — (2)

$\lambda(2x_1 + 5x_2 - 98) = 0$ — (3)

$2x_1 + 5x_2 - 98 = 0$ — (4)

case I : $\lambda = 0$

$4x_1 + 12x_2 = 0$

$-14x_2 + 12x_1 = 0$

$12x_1 - 14x_2 = 0$ $\neq 0$

case II : $\lambda \neq 0$, $2x_1 + 5x_2 = 98$

$4x_1 + 12x_2 - 2\lambda = 0$

$12x_1 - 14x_2 - 5\lambda = 0$

$2x_1 + 6x_2 = \lambda$ from eqn (1)

$12x_1 - 14x_2 - 5(2x_1 + 6x_2) = 0$

$12x_1 - 14x_2 - 10x_1 - 30x_2 = 0$

$2x_1 - 44x_2 = 0$ — (5)

Subtracting eqⁿ (4) & (5)

$$2x_1 - 44x_2 = 0$$
$$-\ 2x_1 + 5x_2 = -98$$
$$-49x_2 = -98$$
$$\boxed{x_2 = 2}$$

Put in eqⁿ (5)

$$2x_1 - 44(2) = 0$$
$$2x_1 - 88 = 0$$
$$\boxed{x_1 = 44}$$

Put x_1, x_2 in $2x_1 + 6x_2 = \lambda$

$$2(44) + 6(2) = \lambda$$
$$88 + 12 = \lambda$$
$$\boxed{\lambda = 100}$$

$$z = 2x_1^2 - 7x_2^2 + 12x_1x_2$$
$$= 2(44)^2 - 7(2)^2 + 12(88)$$
$$\boxed{z = 4900}$$

(B.) Solve the following LPP by Simplex method

Maximize $z = 3x_1 + 5x_2$

Subject to $3x_1 + 2x_2 \le 18$

$0 \le x_1 \le 4$

$0 \le x_2 \le 6$

$x_1, x_2 \ge 0$

$\Rightarrow$ convert to standard form.

$z - 3x_1 - 5x_2 + 0S_1 + 0S_2 + 0S_3 = 0$

Subject to $3x_1 + 5x_2 + S_1 + 0S_2 + 0S_3 = 18$

$x_1 + 0S_1 + S_2 + 0S_3 = 4$

$x_2 + 0S_1 + 0S_2 + S_3 = 6$

CB_j	C_j	3	5	0	0	0		
	BV	x_1	x_2	S_1	S_2	S_3	Soln	Ratio
0	S_1	3	2	1	0	0	18	9
0	S_2	1	0	0	1	0	4	∞
0	S_3	0	1	0	0	1	6	6
	z_j	0	0	0	0	0		
	$c_j - z_j$	3	5	0	0	0		

S_3 leaves and x_2 enters.

CB_j	C_j	3	5	0	0	0		
	BV	x_1	x_2	S_1	S_2	S_3	Soln	Ratio
0	S_1	3	0	1	0	-2	6	2
0	S_2	1	0	0	1	0	4	4
5	x_2	0	1	0	0	1	6	∞
	z_j	0	5	0	0	5		
	$c_j - z_j$	3	0	0	0	-5		

S_1 leaves x_1 enters.

C_{Bj}	C_j	3	5	0	0	0		
	B_V	x_1	x_2	S_1	S_2	S_3	soln	Ratio
3	x_1	1	0	1/3	0	-2/3	2	
0	S_2	0	0	-1/3	1	2/3	2	
5	x_2	0	1	0	0	1	6	
	Z_j	3	5	1	0	3		
	$C_j - Z_j$	0	0	-1	0	-3		

All $c_j - z_j \leq 0$

Optimal solution

$x_1 = 2$, $x_2 = 6$

$Z = 3(2) + 5(6) = 36$

(c) using Big-M method to solve the following LPP

Maximize $z = 3x_1 - x_2$

Subject to $2x_1 + x_2 \geq 2$

$x_1 + 3x_2 \leq 3$

$x_2 \leq 4$

$\Rightarrow$ we introduce the artificial variable A_1 in the 1^{st} constraint and M penalty in the function.

Maximize $z = 3x_1 - x_2 - 0S_1 - 0S_2 - 0S_3 - MA_1$

subject to $2x_1 + x_2 - S_1 + 0S_2 + 0S_3 + A_1 = 2$

$x_1 + 3x_2 + 0S_1 + S_2 + 0S_3 + 0A_1 = 3$

$x_2 + 0S_1 + 0S_2 + S_3 + 0A_1 = 4$

B_j	C_j	3	-1	0	0	0	-M		
	B_x	x_1	x_2	S_1	S_2	S_3	A	soln	Ratio
-M	A_1	2	1	-1	0	0	1	2	1
0	S_2	1	3	0	1	0	0	3	3
0	S_3	0	1	0	0	1	0	4	∞
	z_j	-2M	-M	M	0	0	-M		
	$z_j - c_j$	-2M-3	-M+1	M	0	0	0		

A_1 leaves x_1 enters.

B_j	C_j / BV	3 x_1	-1 x_2	0 S_1	0 S_2	0 S_3	-M A_1	Sol^n	Ratio
3	x_1	1	1/2	-1/2	0	0	1/2	1	2
0	S_2	0	5/2	1/2	1	0	-1/2	2	4/5
0	S_3	0	1	0	0	1	0	4	4
	z_j	3	3/2	-3/2	0	0	3/2		
	$z_j - c_j$	0	5/2	-3/2	0	0	$\frac{3}{2}+M$		

S_2 leaves, x_2 enters

B_j	C_j / BV	3 x_1	-1 x_2	0 S_1	0 S_2	0 S_3	-M A_1	Sol^n	Ratio
3	x_1	1	0	-3/5	-1/5	0	3/5	3/5	
-1	x_2	0	1	1/5	2/5	0	-1/5	4/5	
0	S_3	0	0	-1/5	-2/5	1	1/5	16/5	
	z_j	3	-1	-2	-1	0	2		
	$z_j - c_j$	0	0	-2	-1	0	2+M		

(B) Solve the following LPP by simplex method.

Minimise $z = x_1 - 3x_2 + x_3$

Subject to $3x_1 - x_2 + 2x_3 \leq 7$

$2x_1 + 4x_2 \geq -12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

⇒ convert to standard form

$z - x_1 + 3x_2 - x_3 + 0S_1 + 0S_2 + 0S_3 = 0$

Subject to $3x_1 - x_2 + 2x_3 + S_1 + 0S_2 + 0S_3 = 7$

$-2x_1 - 4x_2 + 0S_1 + S_2 + 0S_3 = 12$

$-4x_1 + 3x_2 + 8x_3 + 0S_1 + 0S_2 + S_3 = 10$

B_j	c_j	1	-3	1	0	0	0		
	BV	x_1	x_2	x_3	S_1	S_2	S_3	soln	Ratio
0	S_1	3	-1	2	1	0	0	7	$7/-1 = -7$
0	S_2	-2	-4	0	0	1	0	12	$12/-4 = -3$
0	S_3	-4	3	8	0	0	1	10	$\frac{10}{3} = 3.33$
	z_j	0	0	0	0	0	0		
	$c_j - z_j$	1	-3	1	0	0	0		

x_2 enter S_3 leave.

B_j	c_j	1	-3	1	0	0	0		
	B_V	x_1	x_2	x_3	S_1	S_2	S_3	Soln	Ratio
0	S_1	5/3	0	14/3	1	0	1/3	31/3	31/5
0	S_2	-22/3	0	32/3	0	1	4/3	76/3	76/-22
-3	x_2	-4/3	1	8/3	0	0	1/3	10/3	10/-4
	z_j	4	-3	-8	0	0	-1		
	c_j-z_j	-3	0	9	0	0	1		

x_1 enters S_1 leaves.

B_j	c_j	1	-3	1	0	0	0		
	B_V	x_1	x_2	x_3	S_1	S_2	S_3	Soln	Ratio
1	x_1	1	0	14/5	3/5	0	1/5	31/5	
0	S_2	0	0	156/5	22/5	1	14/5	354/5	
-3	x_2	0	1	32/5	4/5	0	3/5	58/5	
	z_j	1	-3	-82/5	-3/5	0	-8/5		
	c_j-z_j	0	0	87/5	9/5	0	8/5		

All $c_j - z_j \geq 0$

Hence the optimal solution is

$x_1 = 31/5$, $x_2 = 58/5$, $x_3 = 0$

$$= \left(\frac{31}{5}\right) - 3\left(\frac{58}{5}\right) + 0$$

$$= -143/5$$

(c) using Big-M method to solve the LPP.

Maximize $z = 3x_1 - x_2$

Subject to $2x_1 + x_2 \leq 2$

$x_1 + 3x_2 \geq 3$

$x_2 \leq 4$

$x_1, x_2 \geq 0$

=> we introduce the artificial variable A_2 in the 2nd constraint and M penalty in the maximize function.

maximize $z = 3x_1 - x_2 - 0S_1 - 0S_2 - 0S_3 - MA_2$

subject to $2x_1 + x_2 + S_1 + 0S_2 + 0S_3 + 0A_2 = 2$

$x_1 + 3x_2 + 0S_1 - S_2 + 0S_3 + A_2 = 3$

$x_2 + 0S_1 + 0S_2 + S_3 + 0A_2 = 4$

$x_1, x_2, S_1, S_2, S_3, A_2 \geq 0$

B_j	C_j	3	-1	0	0	0			
	B_V	x_1	x_2	S_1	S_2	S_3	A	Sol	Ratio
0	S_1	2	1	1	0	0	0	2	2
-M	A_2	1	3	0	-1	0	1	3	1
0	S_3	0	1	0	0	1	0	4	4
	z_j	-M	-3M	0	M	0	-M		
	$z_j - c_j$	-M-3	-3M+1	0	M	0	-M		

A_2 leaves x_2 enters

B_j	C_j	3	-1	0	0	0	-M		
	Bv	x_1	x_2	S_1	S_2	S_3	A_2	soln	Ratio
0	S_1	5/3	0	1	1/3	0	-1/3	1	3/5
-1	x_2	1/3	1	0	-1/3	0	1/3	1	3
0	S_3	-1/3	0	0	1/3	1	-1/3	3	-9
	Z_j	-1/3	-1	0	1/3	0	-1/3		
	Z_j-C_j	-10/3	0	0	1/3	0	-1/3+M		

S_1 leaves x_1 enters.

B_j	C_j	3	-1	0	0	0	-M		
	Bv	x_1	x_2	S_1	S_2	S_3	A_2	soln	Ratio
	x_1	1	0	3/5	1/5	0	-1/5	3/5	
	x_2	0	1	-1/5	2/5	0	2/5	4/5	
	S_3	0	0	1/5	-2/5	1	-2/5	16/5	
	Z_j	3	-1	2	1	0	-1		
	Z_j-C_j	0	0	2	1	0	-1+M		

(A) use the bisection Method to find a root of equation $f(x) = x^3 - x - 2$ within interval [1,2]

$\Rightarrow$ $f(x) = x^3 - x - 2$, $a = 1$, $b = 2$

$f(a) = f(1) = (1)^3 - 1 - 2 = -2$

$f(b) = f(2) = (2)^3 - 2 - 2 = 4$

$\therefore f(a) \cdot f(b) < 0$

root lies between 1 and 2.

iteration 1 : $x_0 = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$

$f(x_0) = f(1.5) = (1.5)^3 - 1.5 - 2 = -0.125$

$\therefore f(x_0) \cdot f(b) < 0$

root lies betn 1.5 and 2.

iteration 2 : $x_1 = \frac{a+b}{2} = \frac{1.5+2}{2} = 1.75$

$f(x_1) = f(1.75) = (1.75)^3 - 1.75 - 2 = 1.6093$

$\therefore f(x_1) \cdot f(x_0) < 0$

root lies betn 1.5 and 1.75

iteration 3 : $x_2 = \frac{a+b}{2} = \frac{1.5+1.75}{2} = 1.625$

$f(x_2) = f(1.625) = (1.625)^3 - (1.625) - 2$

$= 0.6660$

$\therefore f(x_2) \cdot f(x_0) < 0$

root lies betn 1.5 and 1.625

iteration 4 : $x_3 = \frac{a+b}{2} = \frac{1.5+1.625}{2} = 1.5625$

$f(x_3) = f(1.5625) = (1.5625)^3 - 1.5625 - 2$

$= 0.2521$

$\therefore f(x_3) \cdot f(x_0) < 0$

root lies betn 1.5 and 1.5625

iteration 5 : $x_5 = \frac{a+b}{2} = \frac{1.5 + 1.5625}{2} = 1.53125$

$f(x) = f(1.53125) = (1.53125)^3 - 1.53125 - 2$

$= 0.05911$

$\therefore f(x_5) \cdot f(x_0) < 0$

root lies between 1.5 and 1.5625.

(B) find the real root of the equation $x^4 - x - 10 = 0$
where $x_0 = 1.6$

$\Rightarrow$ $f(x) = x^4 - x - 10 = 0$

$f'(x) = 4x^3 - 1$

then

$f(x_0) = (1.6)^4 - (1.6) - 10$

$= 6.5536 - 1.6 - 10$

$= -5.0464$

$f'(x_0) = 4(1.6)^3 - 1$

$= 16.384 - 1 = 15.384$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

put $n = 0$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1.6 - \frac{(-5.0464)}{15.384}$$

$= 1.928$

$f(x_1) = (1.928)^4 - (1.928) - 10$

$= 1.8894$

$f'(x_1) = 4(1.928)^3 - 1$

$= 27.6669$

put $n = 1$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.928 - \frac{1.8894}{27.6669}$$

$x_2 = 1.8597$

$f(x_2) = (1.8597)^4 - (1.8597) - 10$
$= 0.1014$

$f'(x_2) = 4(1.928)^3 - 1$
$= 24.7669$

put $n = 2$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.8597 - \frac{0.1014}{24.7669}$$

$x_3 = 1.8554$

$f(x_3) = (1.8554)^4 - (1.8854) - 10$
$= -0.6345$

$f'(x_3) = 4(1.8554)^3 - 1$
$= 24.5489$

(D) Use the secant method to find a root of equation $f(x) = x^3 - x - 2$ where $x_0 = 1$ and $x_1 = 2$

$\Rightarrow$ put $n = 1$

$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$f(x_1) = (x_1)^3 - (x_1) - 2$$
$$= (2)^3 - 2 - 2$$
$$= 8 - 4 = 4$$

$$f(x_0) = (x_0)^3 - (x_0) - 2$$
$$= (1)^3 - 1 - 2$$
$$= 1 - 1 - 2$$
$$f(x_0) = -2$$

$$x_2 = 2 - \frac{4(1-2)}{-2-4}$$
$$= 2 - \frac{-4}{-6} = \frac{12-4}{6}$$
$$= 1.3333$$

$$f(x_2) = (1.3333)^3 - (1.3333) - 2$$
$$= -0.9631$$

put $n = 2$

$$x_3 = x_2 - \frac{f(x_2)(x_1 - x_2)}{f(x_1) - f(x_2)}$$
$$= 1.3333 - \frac{(-0.9631)(2 - 1.3333)}{4 - (-0.9631)}$$
$$= 1.3333 - \frac{(-0.6420)}{4.9631}$$
$$= 1.3333 + 0.1293$$
$$x_3 = 1.4626$$

$$f(x_3) = (1.4626)^3 - 1.4626 - 2$$
$$= -0.3338$$

put $n=3$

$$x_4 = x_3 - \frac{f(x_3)(x_2-x_3)}{f(x_2)-f(x_3)}$$

$$= 1.4626 - \frac{(-0.3338)(1.3333-1.4626)}{-0.9631-(-0.3338)}$$

$$\simeq 1.4626 - \frac{(0.0431)}{-0.6293}$$

$$= 1.4626 + 0.0684$$

$$x_4 = 1.531$$

$$f(x_4) = (1.531)^3 - 1.531 - 2$$

$$= 0.0576$$

put $n=4$

$$x_5 = x_4 - \frac{f(x_4)(x_3-x_4)}{f(x_3)-f(x_4)}$$

$$= 1.531 - \frac{(0.0576)(1.4626-1.531)}{-0.3338-0.0576}$$

$$= 1.531 - \frac{(0.0576)(-0.0684)}{-0.3914}$$

$$= 1.531 - 0.0100$$

$$x_5 = 1.5210$$

$$f(x_5) = (1.5210)^3 - 1.5210 - 2$$

$$= -0.02256$$

(B) Find the real root of the equation $3x - \cos x - 1 = 0$ where $x_0 = 0.6$

$\Rightarrow$ $\because$ $f(x) = 3x - \cos x - 1$

$f'(x) = 3 + \sin x$

$x_0 = 0.6$ is given

$f(x_0) = 3(0.6) - \cos(0.6) - 1$

$= -0.19995$

$f'(x_0) = 3 + \sin(0.6)$

$= 3.01047$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

put $n = 0$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.6 - \frac{(-0.19995)}{3.01047}$$

$= 0.6 + 0.0664$

$x_1 = 0.6664$

$f(x_1) = 3(0.6664) - \cos(0.6664) - 1$

$= 0.0007323$

$f'(x_1) = 3 + \sin(0.6664)$

$= 3.0116$

put $n = 1$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.6664 - \frac{0.0007323}{3.0116}$$

$x_2 = 0.6661$

(c) Solve the root of $f(x) = x^3 + x^2 - 3x - 3$
where $x_0 = 1$, $x_1 = 2$ formula $\{ x_{n+1} = x_n - \frac{f(x_n)(x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$

$\Rightarrow$ put $n = 1$

$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$f(x_1) = (x_1)^3 + (x_1)^2 - 3(x_1) - 3$$
$$= 2^3 + 2^2 - 3(2) - 3$$
$$= 3$$

$$f(x_0) = (x_0)^3 + (x_0)^2 - 3(x_0) - 3$$
$$= (1)^2 + 1^2 - 3 - 3$$
$$= -4$$

$$x_2 = 2 - \frac{3(1-2)}{-4-3}$$
$$= 2 - \frac{(-3)}{-7} = 2 - \frac{3}{7} = \frac{11}{7}$$

$$x_2 = 1.5714$$

$$f(x_2) = (1.5714)^3 + (1.5714)^2 - 3(1.5714) - 3$$
$$= -1.3646$$

put $n = 2$

$$x_3 = x_2 - \frac{f(x_2)(x_1 - x_2)}{f(x_1) - f(x_2)}$$
$$= 1.5714 - \frac{(-1.3646)(0.4286)}{3 - (-1.3646)}$$
$$= 1.5714 + \frac{0.5848}{4.3646}$$

$$x_3 = 1.7053$$

$$f(x_3) = (1.7053)^3 + (1.7053)^2 - 3(1.7053) - 3$$
$$= -0.2487$$

put $n=3$

$$x_4 = x_3 - \frac{f(x_3)(x_2-x_3)}{f(x_2)-f(x_3)}$$

$$= 1.7053 - \frac{(-0.2487)(-0.1339)}{-1.3646+0.2487}$$

$$x_4 = 1.7351$$

$$f(x_4) = (1.7351)^3 + (1.7351)^2 - 3(1.7351) - 3$$

$$= 0.0289$$

put $n=4$

$$x_5 = x_4 - \frac{f(x_4)(x_3-x_4)}{f(x_3)-f(x_4)}$$

$$= 1.7351 - \frac{(0.0289)(-0.0298)}{-0.2776}$$

$$= 1.7351 - 0.00310$$

$$x_5 = 1.732$$

$$f(x_5) = (1.732)^3 + (1.732)^2 - 3(1.732) - 3$$

$$= 0.00048$$

(D) Use the Bisection Method to find root of the equation $f(x) = x^2 - 3$ within interval $[1, 2]$

$\Rightarrow$ $f(x) = x^2 - 3$, $a = 1$, $b = 2$

$f(a) = f(1) = (1)^2 - 3 = -2$

$f(b) = f(2) = (2)^2 - 3 = 1$

$\therefore f(a) . f(b) < 0$

root lies betn 1 and 2.

iteration 1 : $x_0 = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$

$f(x_0) = f(1.5) = (1.5)^2 - 3 = -0.75$

$\therefore f(x_0) . f(b) < 0$

root lies between 1.5 and 2.

iteration 2 : $x_1 = \frac{a+b}{2} = \frac{1.5+2}{2} = 1.75$

$f(x_1) = f(1.75) = (1.75)^2 - 3 = 0.0625$

$\therefore f(x_0) . f(x_1) < 0$

root lies betn 1.5 and 1.75

iteration 3 : $x_2 = \frac{a+b}{2} = \frac{1.5+1.75}{2} = 1.625$

$f(x_2) = f(1.625) = (1.625)^2 - 3 = 0.3594$

$f(x_2) . f(x_1) < 0$

$\therefore$ root lies betn 1.75 and 1.625

iteration 4 : $x_3 = \frac{a+b}{2} = \frac{1.625+1.75}{2} = 1.6875$

$f(x_3) = f(1.6875) = (1.6875)^2 - 3 = -0.1523$

$f(x_3) . f(x_1) < 0$

root lies betn 1.6875 and 1.75

iteration 5 : $x_4 = \frac{a+b}{2} = \frac{1.6875+1.75}{2} = 1.71875$

$f(x_4) = f(1.71875) = (1.71875)^2 - 3 = -0.045898$

$f(x_4) \cdot f(x_1) < 0$

root lies betn 1.71875 and 1.75.

(c) Use the Golden Search method to minimize the function $f(x) = (x-2)^2$ within the interval $[0,4]$.

$$d = \frac{\sqrt{5}-1}{2}(4-0)$$

$$= \sqrt{5}-1 \ (2)$$

$$d = 2.4721$$

$x_1 = x_l + 2.4721$ $\therefore x_1 = 2.4721$

$x_2 = x_u - 2.4721$ $\therefore x_2 = 1.5279$

$$f(x_1) = (2.4721-2)^2$$
$$= 0.2228$$

$$f(x_2) = (1.5279-2)^2$$
$$= 0.2228$$

Since both are equal $x_l = 1.5280$, $x_u = 2.4721$

$$d = \frac{\sqrt{5}-1}{2}(2.472-1.528)$$

$$d = 0.5834$$

$x_1 = 0.5834 + 1.528$ $\therefore x_1 = 2.1114$

$x_2 = 2.472 - 0.5834$ $\therefore x_2 = 1.8886$

$f(x_1) = (2.1114-2)^2 = 0.0124$

$f(x_2) = (1.8886-2)^2 = 0.0124$

Since both are equal $x_1 = 1.8886$, $x_2 = 2.1114$

$$d = \frac{\sqrt{5}-1}{2}(2.1114-1.8886)$$

$$d = 0.1376$$

$x_1 = 1.8886 + 0.1376$ $\qquad \therefore x_1 = 2.0263$

$x_2 = 2.1114 - 0.1376$ $\qquad \therefore x_2 = 1.9738$

$f(x_1) = (2.0263 - 2)^2$

$= 0.0006$

$f(x_2) = (1.9738 - 2)^2$

$= 0.0006$

At this point, the interval length is small (0.0525) and clearly the function is minimum

PAGE No.
DATE / /

* Maximize $f(\theta) = 4\sin\theta(1+\cos\theta)$, $[0, \frac{\pi}{2}]$

Iteration 1 :

$$x_l = 0 \quad , \quad x_u = \frac{\pi}{2}$$

$$x_1 = x_l + \frac{\sqrt{5}-1}{2}(x_u - x_l)$$

$$= 0 + \frac{\sqrt{5}-1}{2}(1.5708)$$

$$x_1 = 0.9708$$

$$x_2 = x_u - \frac{\sqrt{5}-1}{2}(x_u - x_l)$$

$$x_2 = 0.6000$$

$$f(x_1) = 4\sin(0.9708)(1+\cos(0.9708))$$
$$= 0.0677(1.9998)$$
$$= 0.1353$$

$$f(x_2) = 4\sin(0.6000)(1+\cos(0.6000))$$
$$= 0.0837$$

Since $f(x_1) > f(x_2)$.

$$x_1 = x_l + \frac{\sqrt{5}-1}{2}(x_u - x_l)$$

$$= 0.6000 + \frac{\sqrt{5}-1}{2}(1.5708 - 0.6000)$$

$$= 1.2000$$